



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

CASE STUDY-04

Practical 4: ATM Card PIN Verification Using JavaScript Functions

1.Aim:

To verify whether an ATM PIN is a palindrome using JavaScript functions and demonstrate function declaration, function expression, arrow function, and scope.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ATM PIN Verification</title>
```

```
<style>
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial, Helvetica, sans-serif;
}

body{
  height:100vh;
  display:flex;
  justify-content:center;
  align-items:center;
  background:linear-gradient(135deg,#0f2027,#203a43,#2c5364);
}
```

```
.container{
  width:400px;
  background:white;
  padding:30px;
  border-radius:20px;
  text-align:center;
  box-shadow:0 10px 30px rgba(0,0,0,0.4);
}
```



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}

```
button:hover{
  background:#0d47a1;
  transform:scale(1.05);
}
```

```
#output{
  margin-top:25px;
  padding:15px;
  border-radius:10px;
  display:none;
}
```

```
.success{
  background:#d4edda;
  color:#155724;
  border:2px solid #28a745;
}
```

```
.error{
  background:#f8d7da;
  color:#721c24;
  border:2px solid #dc3545;
}
```

```
.footer{
  margin-top:20px;
  color:#666;
  font-size:14px;
}
```

</style>

</head>

<body>

<div class="container">

<div class="icon">  </div>



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<h2>ATM PIN Verification</h2>

<p>Enter your 4-digit ATM PIN</p>

<input type="password" id="pin" placeholder="Enter PIN">

<button onclick="checkPIN()">Verify PIN</button>

<div id="output"></div>

<div class="footer">

ABC Bank Security System

</div>

</div>

<script>

// Global Variable

let bankName="ABC Bank";

// Function Declaration

```
function reversePIN(pin){  
  return pin.split("").reverse().join("");  
}
```

// Function Expression

```
const isPalindrome=function(pin){  
  return pin===reversePIN(pin);  
}
```

// Arrow Function

```
const displayResult=(pin)=>{
```

```
  let output=document.getElementById("output");
```

```
  if(isPalindrome(pin)){
```

```
    output.className="success";
```

```
    output.innerHTML=
```

Experiment No.: 4

1. Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

Function Types: Demonstrates anonymous functions (assigned directly to variables without a name) and arrow functions (concise ES6 => syntax).

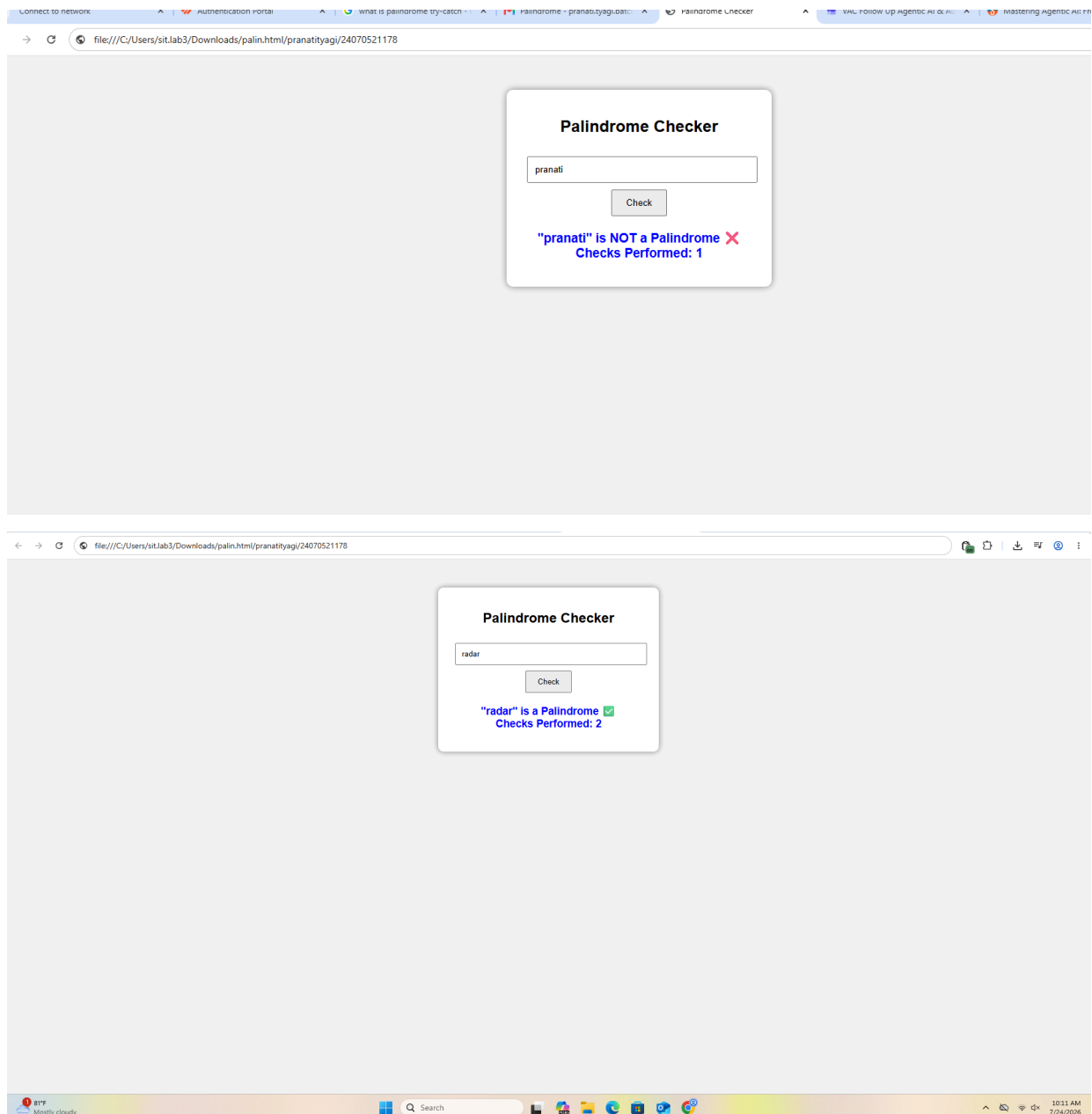
Scope: Differentiates between global variables (accessible throughout the script) and local variables (restricted to the block or function where they are declared).

Closures: Occurs when an inner function retains access to its parent function's scope, preserving state variables (like count) even after the outer function finishes executing.

Error Handling: Uses try...catch and throw statements to validate input types and gracefully handle runtime exceptions without crashing the program.

Palindrome Algorithm: A string is split into an array of characters, reversed, joined back into a string, and compared against the original input using strict equality (===).

4. Output:



5. Result/Conclusion: In this experiment, fundamental JavaScript concepts—including anonymous/arrow functions, scope rules, closures, and try...catch error handling—were successfully implemented. Using these techniques alongside array manipulation methods (split, reverse, join), a functional and error-resistant palindrome checker was constructed.

